



# Manufacturing Engineers

**Also called:** Manufacturing Engineer, Plant Engineer, Process Engineer, Process Improvement Engineer

## What they do:

Design, integrate, or improve manufacturing systems or related processes. May work with commercial or industrial designers to refine product designs to increase producibility and decrease costs.

## On the job, you would:

- Troubleshoot new or existing product problems involving designs, materials, or processes.
- Investigate or resolve operational problems, such as material use variances or bottlenecks.
- Identify opportunities or implement changes to improve manufacturing processes or products or to reduce costs, using knowledge of fabrication processes, tooling and production equipment, assembly methods, quality control standards, or product design, materials and parts.

## KNOWLEDGE

### Engineering and Technology

- product and service development
- mechanical

### Manufactured or Agricultural Goods

- manufacture and distribution of products

### Math and Science

- arithmetic, algebra, geometry, calculus, or statistics
- physics

### Arts and Humanities

- English language

## SKILLS

### Basic Skills

- reading work related information
- using math to solve problems

### Problem Solving

- noticing a problem and figuring out the best way to solve it

### People and Technology Systems

- thinking about the pros and cons of different options and picking the best one
- figuring out how a system should work and how changes in the future will affect it

## ABILITIES

### Ideas and Logic

- group things in different ways
- make general rules or come up with answers from lots of detailed information

### Verbal

- listen and understand what people say
- communicate by speaking

### Math

- add, subtract, multiply, or divide
- choose the right type of math to solve a problem

### Visual Understanding

- see hidden patterns

## PERSONALITY

People interested in this work like activities that include **practical, hands-on** problems and solutions.

They do well at jobs that need:

- **Innovation**
- **Adaptability**
- **Intellectual Curiosity**
- **Attention to Detail**
- **Dependability**
- **Achievement Orientation**

## TECHNOLOGY

You might use software like this on the job:

### Computer aided design CAD software

- Autodesk AutoCAD 🔥 📈
- Dassault Systemes SolidWorks 🔥 📈

### Desktop communications software

- Eko

### Electronic mail software

- IBM Notes
- Microsoft Outlook 🔥

## EDUCATION



**bachelor's degree** or  
**associate's degree or other 2-year degree**

usually needed

## JOB OUTLOOK

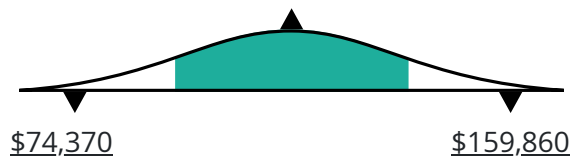


New job opportunities are **very likely** in the future.

**Bright**

SALARY:

**\$102,440**



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